

HANDSMEN THREADS Salesforce-Based Clothing Management System

ABSTRACT

HandsMen Threads is a Salesforce-based clothing management system designed to manage customers, products, orders, and inventory in a centralized platform. The project uses Salesforce Flows, Apex Triggers, Batch Apex, Scheduled Apex, and Email Alerts to automate key business processes. It provides automated customer loyalty management, order validation, low-stock notifications, and inventory processing, reducing manual effort and improving the efficiency and accuracy of business operations.

1. HandsMen Threads Salesforce Application

1.1 Application Overview

The HandsMen Threads application is a Salesforce-based management system developed to manage the major operations of a clothing business. The application provides a centralized platform for managing customers, orders, products, and inventory. Salesforce automation features such as Flows, Apex Trigger, Batch Apex, Scheduled Apex, and Email Alerts are integrated into the application to automate business processes and reduce manual effort.

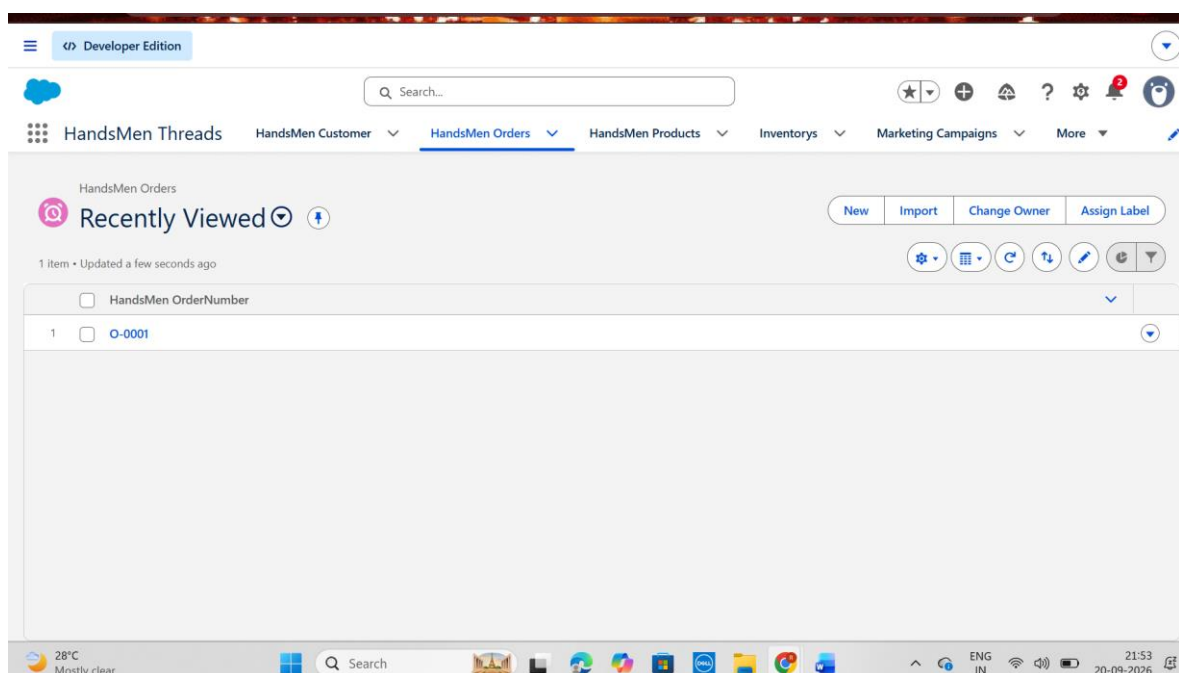


Figure 1.1: HandsMen Threads Salesforce Application

2. Customer Management

2.1 Customer Record Management

The HandsMen Customer object is used to store and manage customer information. It maintains important details such as customer name, phone number, total purchases, and loyalty status. The customer information is also used by the loyalty automation to determine the appropriate loyalty level.

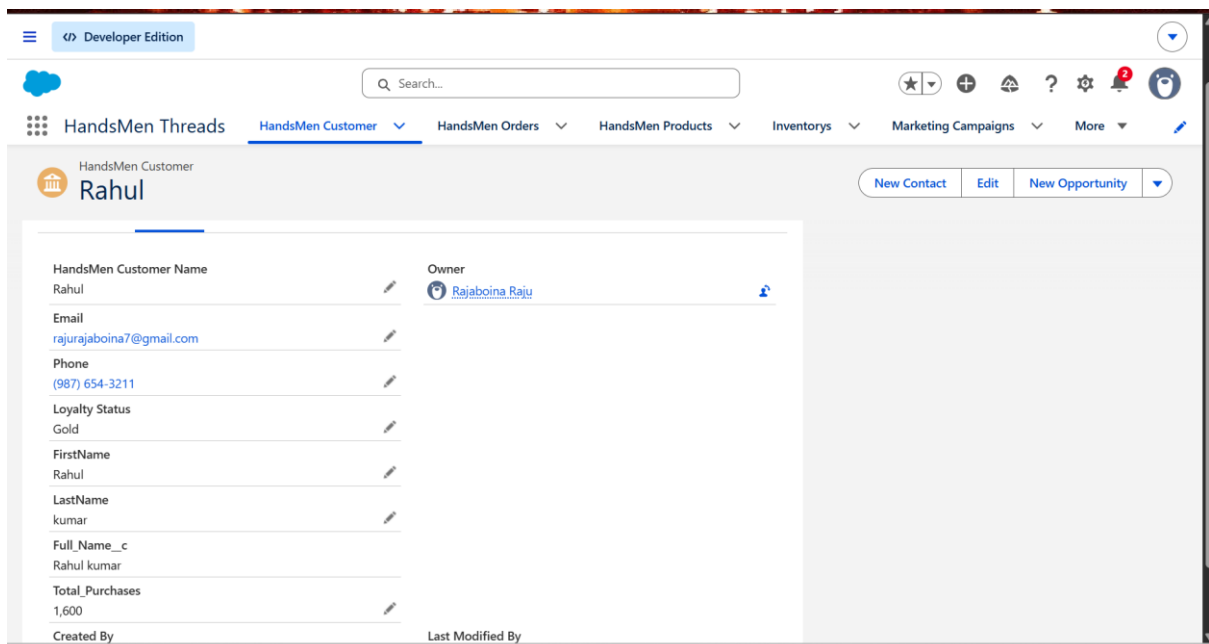


Figure 2.1: HandsMen Customer Record

3. Loyalty Management

3.1 Automated Loyalty Status Management

The project implements an automated loyalty management system using a Scheduled-Triggered Flow. The Flow evaluates the total purchases of each customer and automatically assigns the appropriate loyalty status.

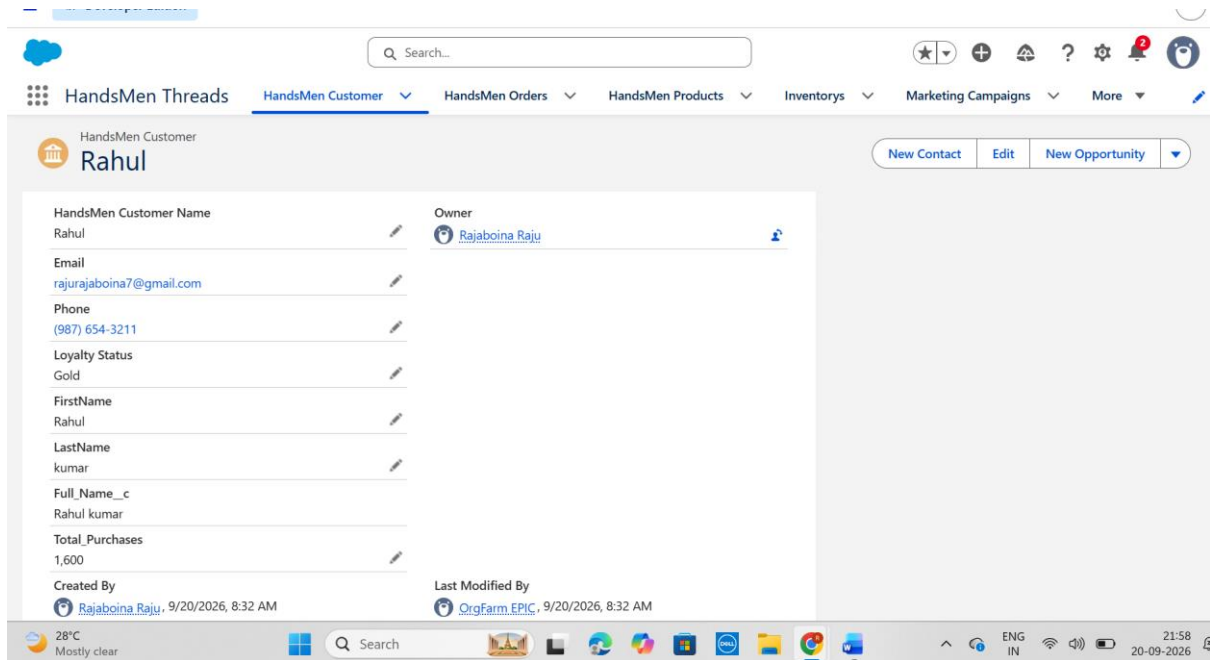


Figure 3.1: Customer Loyalty Status

3.2 Loyalty Status Update Flow

The Loyalty Status Update Flow retrieves customer records, processes them using a loop, evaluates the purchase amount through a Decision element, and updates the corresponding loyalty status.

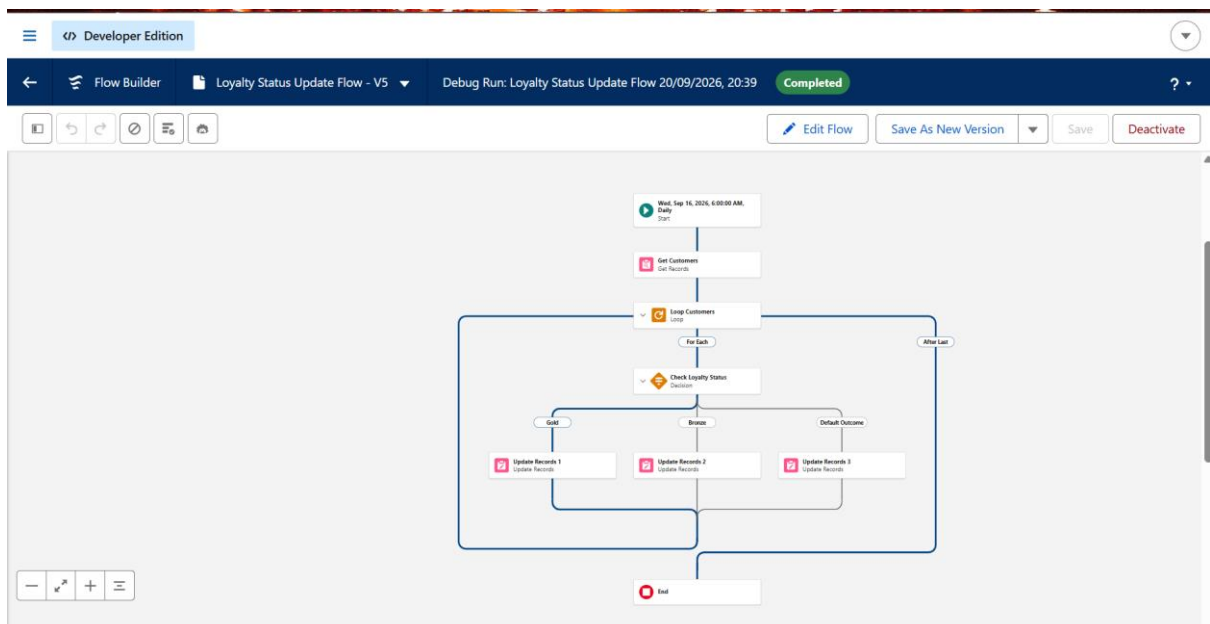


Figure 3.2: Loyalty Status Update Flow

4. Order Management

4.1 Order Management

The HandsMen Order object is used to maintain customer order information. Important details include the associated customer, order quantity, order status, and total amount.

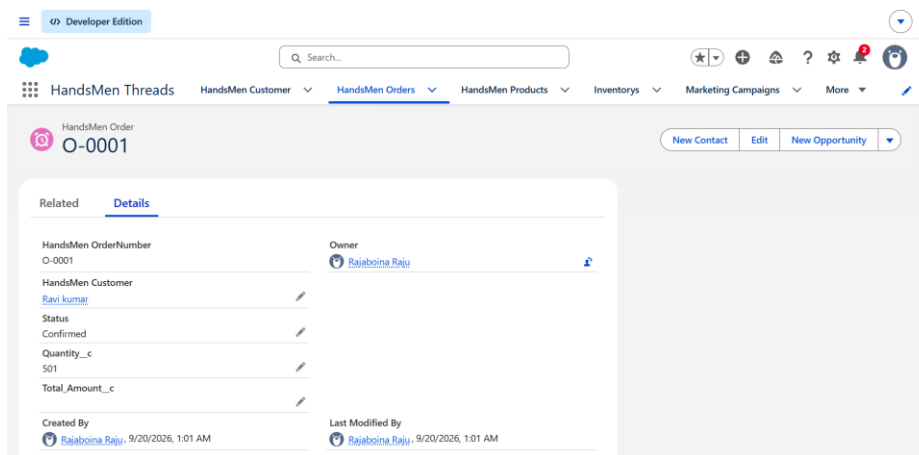


Figure 4.1: HandsMen Order Record

5. Order Validation Using Apex

5.1 Order Validation Trigger

An Apex Trigger is implemented to validate order quantities according to the order status. The trigger executes before insertion and before updating an order record.

```
1 public class OrderTriggerHandler {
2     public static void validateOrderQuantity(List<HandsMen_Order__c> orderList) {
3
4         for (HandsMen_Order__c order : orderList) {
5
6             if (order.Status__c == 'Confirmed') {
7
8                 if (order.Quantity_c == null || order.Quantity_c <= 500) {
9
10                    order.Quantity_c__c.addError(
11                        'For Status "Confirmed", Quantity must be more than 500.'
12                    );
13                }
14            }
15        }
16    }
17    } else if (order.Status__c == 'Pending') {
18
19        if (order.Quantity_c == null || order.Quantity_c <= 200) {
20
21            order.Quantity_c__c.addError(
22                'For Status "Pending", Quantity must be more than 200.'
23            );
24        }
25    }
26    } else if (order.Status__c == 'Rejection') {
27
28        if (order.Quantity_c == null || order.Quantity_c <= 0) {
29
30            order.Quantity_c__c.addError(
31                'For Status "Rejection", Quantity must be 0.'
32            );
33        }
34    }
35    }
36    }
37    }
38    }
39    System.debug('All records validated successfully.');
```

Figure 5.1: Order Trigger Implementation

6. Product Management

6.1 Product Management

The HandsMen Product object stores product-related information such as product name, SKU, price, and stock quantity. Products can be associated with orders and inventory records.

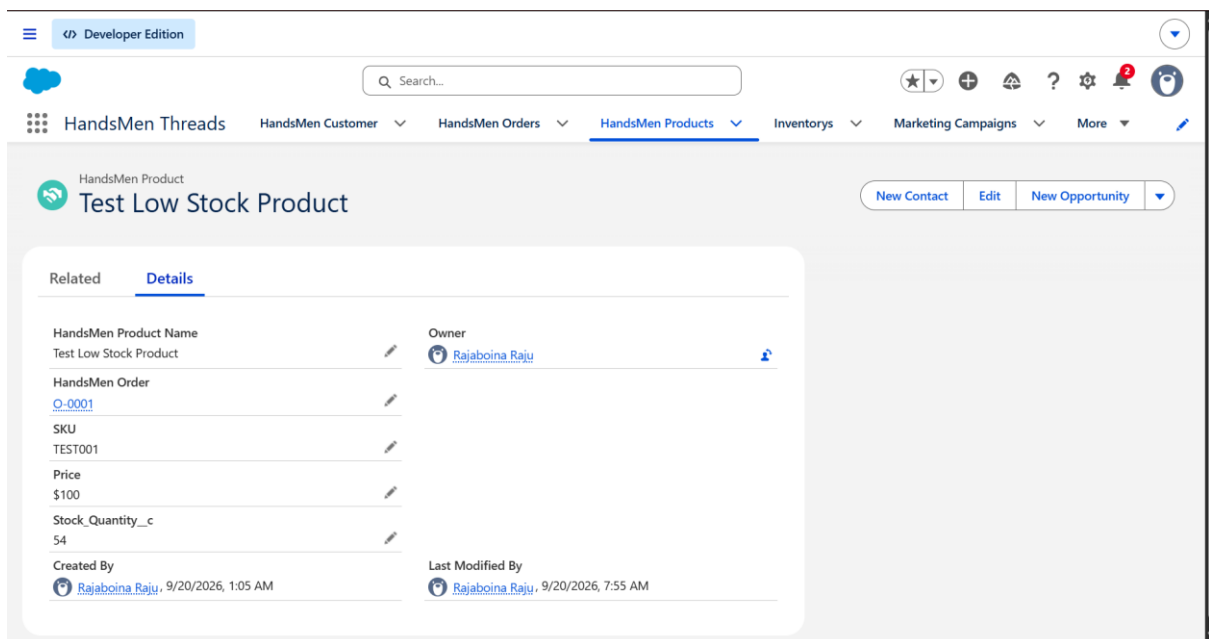


Figure 6.1: HandsMen Product Record

7. Inventory Management

7.1 Inventory Management

The Inventory object is used to maintain stock information for products. It contains the associated product, warehouse information, and current stock quantity.

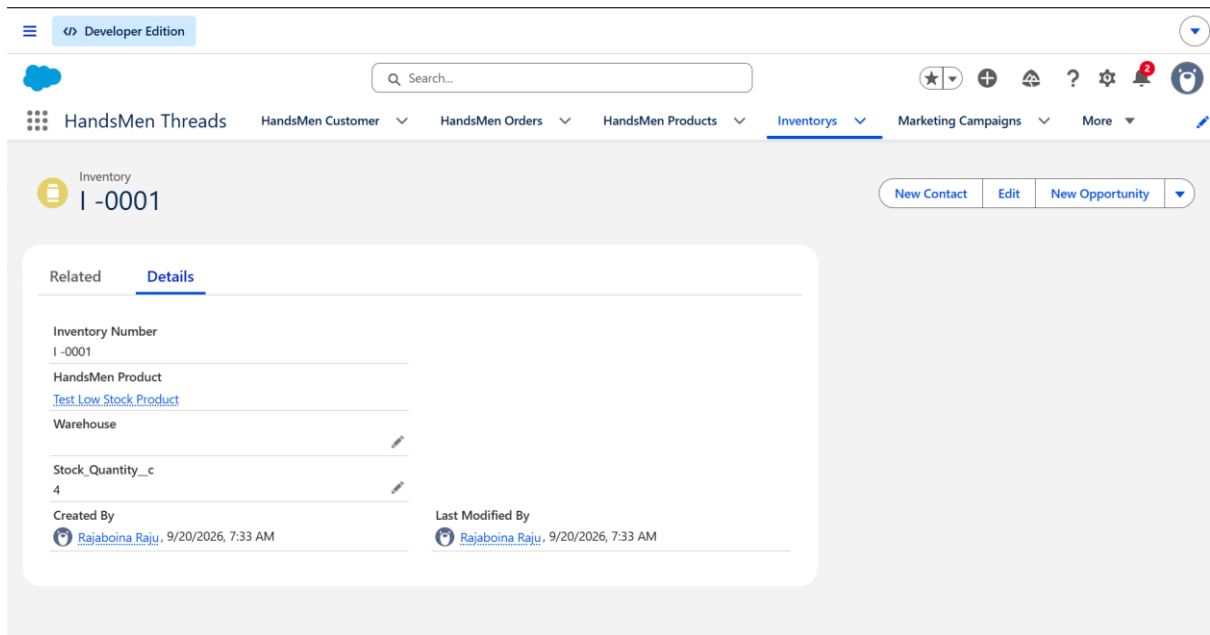


Figure 7.1: Inventory Record

8. Low Stock Alert Automation

8.1 Low Stock Detection

A Record-Triggered Flow is implemented on the Inventory object to detect low-stock conditions. The Flow executes when an inventory record is created or updated. When the stock quantity falls below 5, the Flow triggers the configured Low Stock Email Alert.

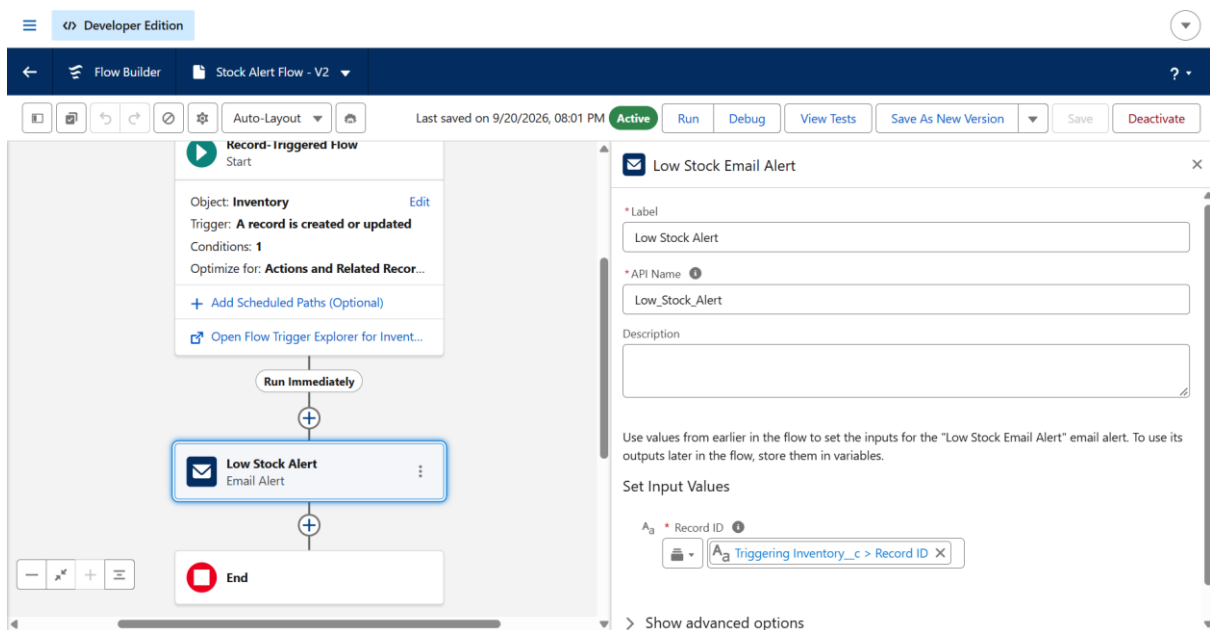


Figure 8.1: Low Stock Alert Flow

9. Low Stock Email Alert

9.1 Email Notification

An Email Alert is configured for the Inventory object using the Low Stock Alert email template. When the low-stock condition is satisfied, the notification is automatically sent to the configured recipient.

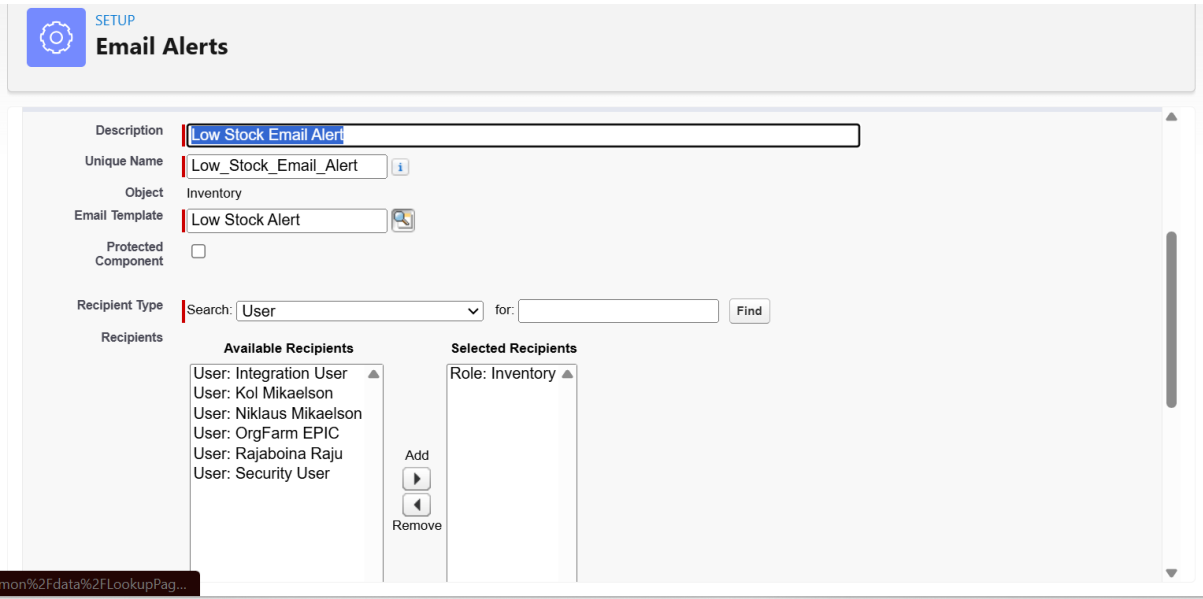


Figure 9.1: Low Stock Email Alert Configuration

9.2 Generated Email

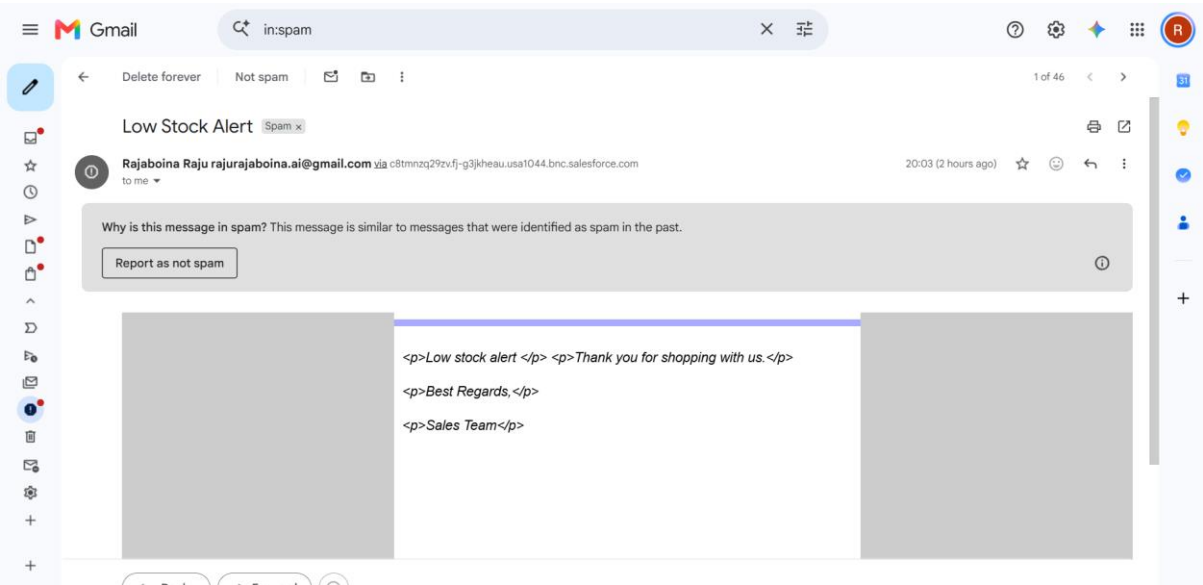


Figure 9.2: Low Stock Email Notification

10. Inventory Batch Processing

10.1 Batch Apex Implementation

The project uses Batch Apex to process low-stock products in batches. The InventoryBatchJob class implements the Batchable interface and performs the required inventory processing.

Batch Apex provides a scalable way to process a large number of Salesforce records while respecting platform governor limits.

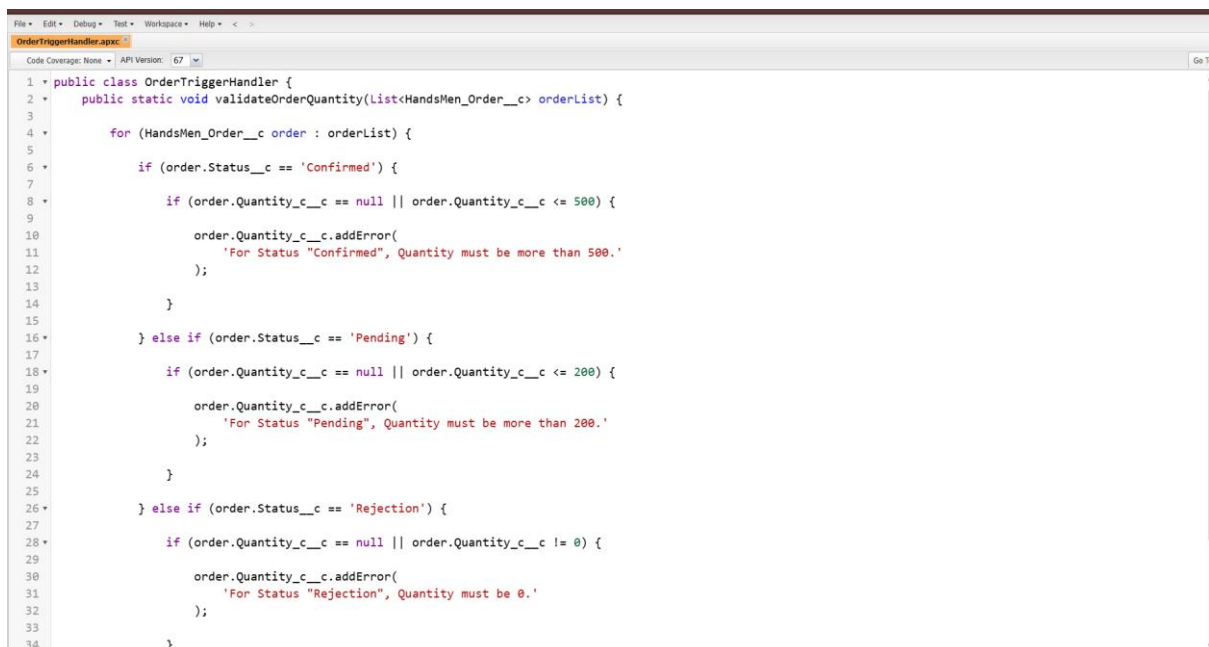


Figure 10.1: Inventory Batch Apex Implementation

11. Scheduled Inventory Processing

11.1 Scheduled Apex

The Inventory Batch Job is also implemented as a Schedulable Apex class. This allows the inventory processing operation to be executed automatically according to a configured schedule.

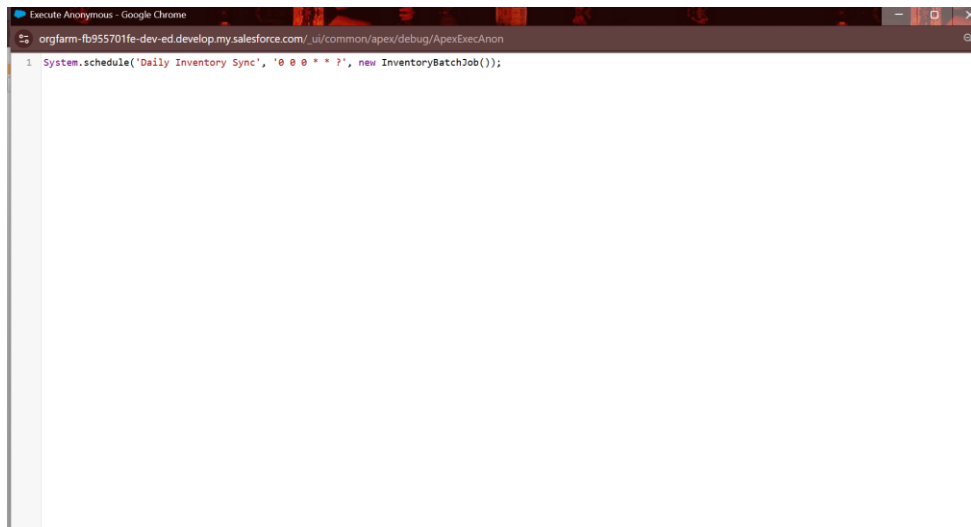


Figure 11.1: Inventory Batch Scheduler

12. Email Templates

12.1 Classic Email Templates

Two Classic Email Templates were created as part of the project:

- Low Stock Alert
- Loyalty Program Email

These templates are used to support automated communication within the system.

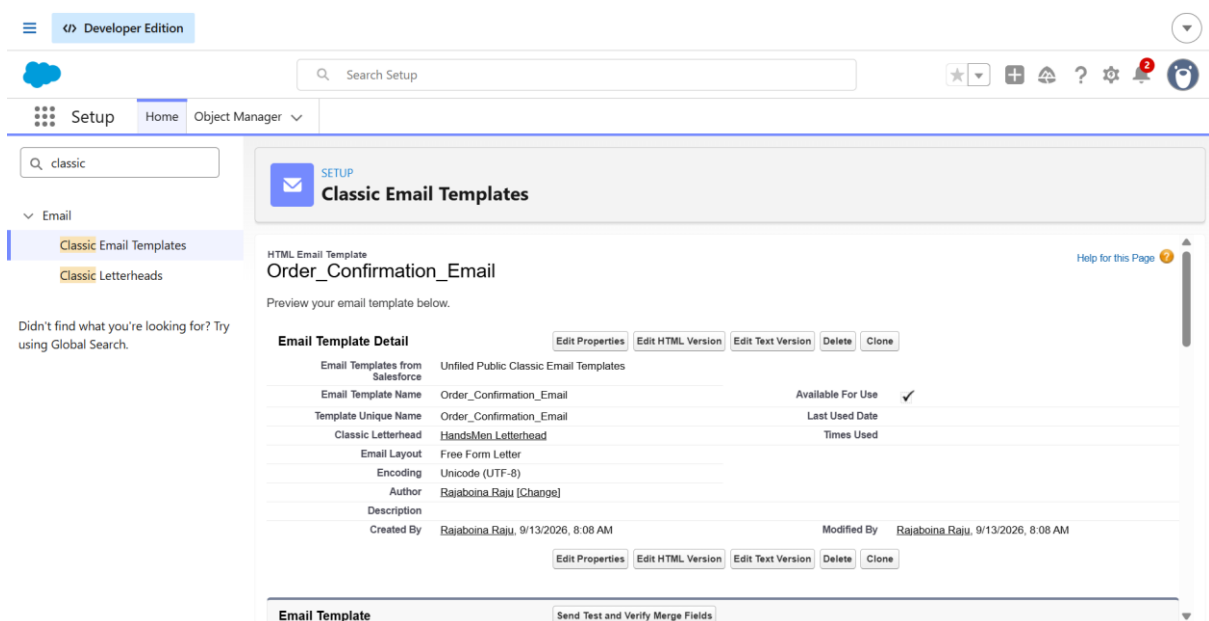


Figure 12.1: HandsMen Threads Email Templates

14. Conclusion

The HandsMen Threads Salesforce application successfully integrates Salesforce objects, automation, and Apex programming to manage clothing-business operations. The system provides centralized customer, product, order, and inventory management while automating loyalty status updates, order validation, low-stock notifications, and inventory processing.

The project demonstrates the practical use of Salesforce Flows, Email Alerts, Apex Trigger, Batch Apex, and Scheduled Apex to implement real-world business requirements.